3 Hydroformylation shown in Fig. 3.1 is one of the largest homogeneous catalytic reactions used to manufacture aldehydes industrially. The aldehydes are subsequently hydrogenated to alcohols or oxidised to carboxylic acids.

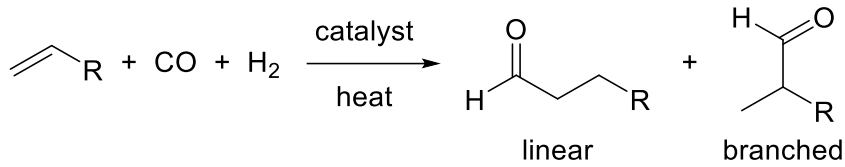


Fig. 3.1

The phosphine-modified cobalt catalyst, $HCo(CO)_3(PR_3)$, was developed to increase the linear : branched aldehyde selectivity. This catalyst also functions as an effective aldehyde hydrogenation catalyst to produce the corresponding alcohol.

The proposed mechanisms for the hydroformylation cycle are shown in Fig. 3.2.

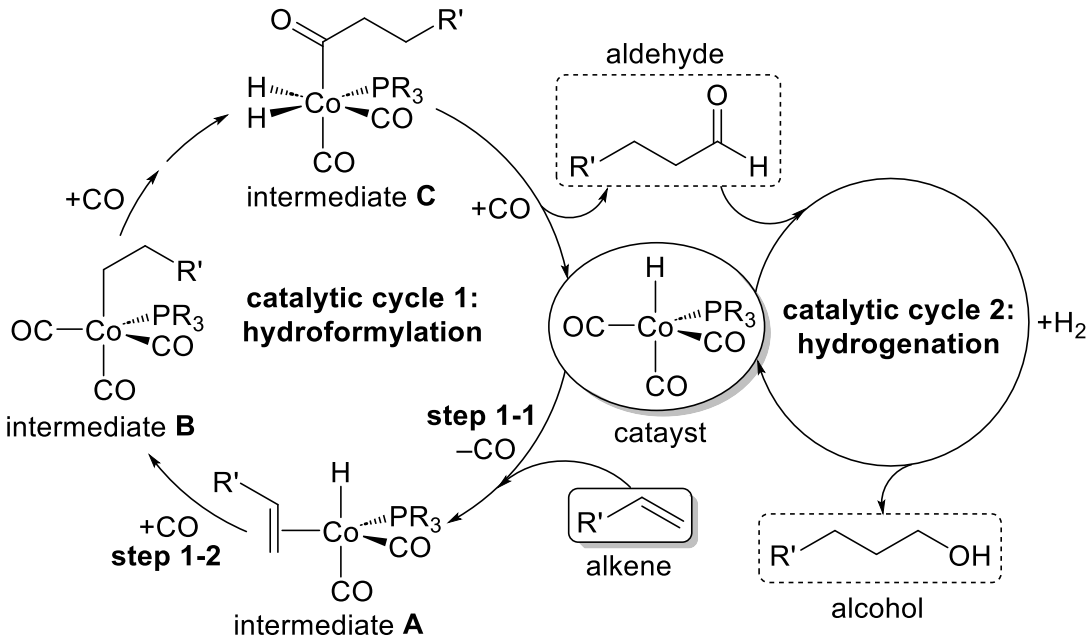


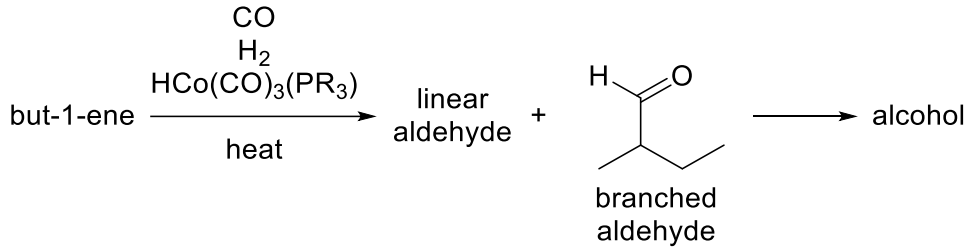
Fig. 3.2

An important step in catalytic cycle 1 is **step 1-2**, which involves addition of a Co–H bond adds across the C=C double bond within intermediate **A**, with binding of a carbon monoxide ligand, to give intermediate **B**.

Intermediate **B** reacts with more carbon monoxide, eventually giving intermediate **C**. Finally, intermediate **C** breaks down to give the aldehyde with regeneration of the catalyst.

The aldehyde formed from catalytic cycle 1 then enters catalytic cycle 2 and is reduced to give the corresponding alcohol.

(a) But-1-ene was reacted with CO and H_2 using $HCo(CO)_3(PR_3)$ as catalyst.



(i) Draw the skeletal structure of the linear aldehyde. [1]

(ii) Starting from but-1-ene, suggest an alternative 5-step synthesis to form the branched aldehyde. The synthesis involves a nitrile intermediate. [5]

(iii) Suggest a simple chemical test to confirm that all the aldehydes have been hydrogenated to the corresponding alcohols. [2]

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(b) **(i)** State the type of reaction in **step 1-1**. [1]

(ii) State the type of bond between the alkene and Co in intermediate **A** and suggest how it is formed. [2]

(iii) State the shapes of intermediate **B** and intermediate **C** about Co. [1]

(iv) Given that the Co in intermediate **C** is in the +3 oxidation state. State its full electronic configuration. [1]

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(c) The linear and branched product arises in **step 1-2** as the addition of Co–H across the C=C double bond can occur in two ways as shown in Fig. 3.3.

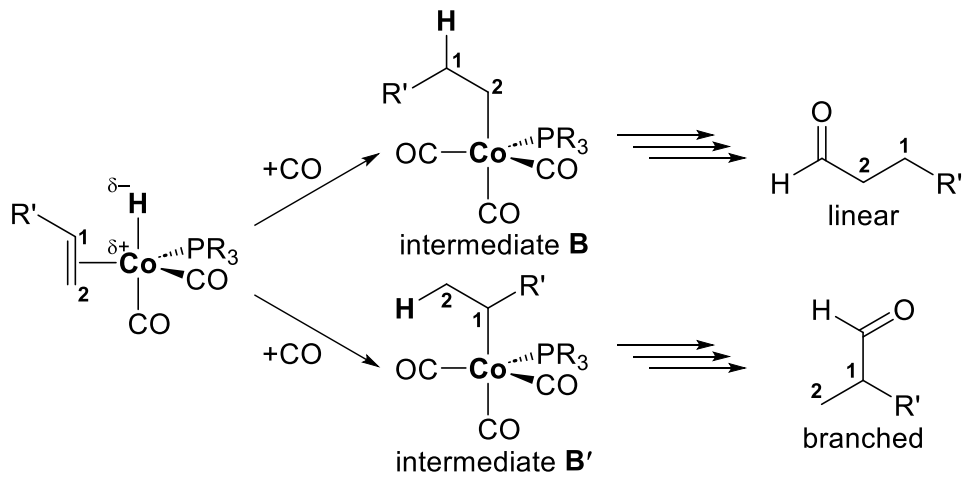

Fig. 3.3

(i) Based on your understanding of the mechanism of electrophilic addition, explain why intermediate **B** is formed preferentially over intermediate **B′**. [2]

(ii) Phosphine ligand, PR_3, with different R groups gives different linear : branched product ratio. State and explain one characteristic of the alkyl group, R, which will further favour formation of intermediate **B**. [2]

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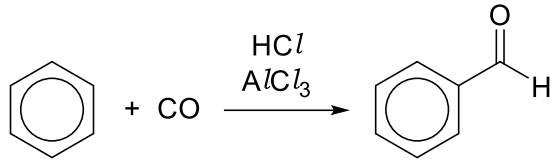
The Friedel-Crafts formylation of benzene using carbon monoxide can be achieved by the Gatterman-Koch reaction in which CO(g) and HC*l*(g) are used *in-situ* with a Lewis acid catalyst such as A*l*C*l*$_3$.



Assuming that CO(g) and HC*l*(g) reacts in the presence of A*l*C*l*$_3$ as shown.

$$CO + HCl + AlCl_3 \rightleftharpoons {}^{+}CHO + AlCl_4^{-}$$

(d) Describe the mechanism for the Friedel-Crafts formylation of benzene to give benzaldehyde. Show the displayed structure of the electrophile, the structure of the intermediate and the movement of electron pairs by using curly arrows. [3]

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[Total: 20]